Scalability & Future Plan

Date	05-07-2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Supports only crop recommendation	Does not provide fertilizer or irrigation suggestions	High
2	Uses a static dataset	Prediction accuracy depends on the available dataset	Medium
3	Runs only on a local Flask server	Users cannot access the application online	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a single/local user	Deploy on cloud (AWS, Render, Azure) to support multiple users
2	Data Storage	Static CSV dataset	Integrate MySQL or MongoDB database
3	Performance	Local ML prediction	Optimize model and use cloud-based APIs for faster response
4	Security	Basic local application	Add user authentication, HTTPS, and secure API access

Future Roadmap:

Phase	**Planned Feature / Enhancement**	**Target Timeline**	**Expected Impact**
Phase 2	Add Fertilizer Recommendation System	Next 2 Months	Helps farmers improve crop productivity
Phase 3	Integrate Weather Forecast and Soil Sensor Data	Next 4 Months	Provides more accurate crop recommendations
Phase 4	Deploy the application on the Cloud with Mobile Support	Next 6 Months	Enables access for farmers from anywhere using mobile devices